

Weak lensing of the DESI DR1 Lyman- α forest by low-redshift galaxies and quasars: estimator, validation and a first amplitude constraint

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Abstract

Gravitational lensing by structure at $z < 2$ displaces the Lyman- α forest sightlines coherently, which dilates and shears the transverse correlation of the forest flux. We measure this effect with a pair-based quadratic estimator: the products of forest pixels on pairs of sightlines are weighted by the response of the measured forest correlation to a deflection and projected on a template of the lensing deflection built from DESI DR1 galaxies and quasars at $0.4 < z < 1.75$ (LRG, ELG and QSO samples in five redshift slices). The tracer template shares no density modes with the forest at $2.1 < z < 3$, so unlike a CMB-lensing template it is free of the intrinsic forest–density three-point term that we found to bias the CMB cross-correlation at the 0.4 level of the lensing amplitude. On 400 self-lensed mock realisations with lognormal tracers the estimator is unbiased (null -0.05 ± 0.08 , recovery 0.91 ± 0.08 , paired response 0.966 ± 0.025 for a true amplitude of one). Applied to 373 756 DR1 forests over $11\,000\,\text{deg}^2$ and templates whose linear biases are measured from their own angular power spectra and cross-checked against ACT DR6 CMB lensing, the lensing amplitude relative to the Λ CDM prediction is $A = 0.01 \pm 0.67$ (single slab), i.e. $A < 1.1$ at 95 per cent, with curl, random-template and injection tests consistent with no systematic. A detection at this depth was not expected; the full DESI sample should reach $2\text{--}3\sigma$.

1 Summary

This note documents the measurement in a self-contained way: the data (Section 2), the estimator and its templates (Section 3), the mock validation that defines its acceptance (Section 4), the DR1 result with its null tests (Section 5) and the caveats (Section 6). The derivations of the estimator and the CMB-lensing forecasts are in the project’s main report (`report/main.pdf`); the pipeline is `code/pipeline` (mocks, estimator, campaign) and `code/stageb` (DR1). Every number quoted here is in the committed JSON products under `report/`.

2 Data

Forest. The DESI DR1 Lyman- α forest deltas [??]: 428 403 forests extracted with `picca v9` over $1040\text{--}1205\,\text{\AA}$ rest frame on a $0.8\,\text{\AA}$ grid, with the lines, DLA (combined CNN + Gaussian-process catalogue, $N_{\text{HI}} > 10^{20.3}\,\text{cm}^{-2}$) and BAL masks applied by the DR1 extraction. We keep pixels with $2.1 \leq z \leq 3.0$ and forests with at least 50 such pixels: 373 756 forests, 1.8×10^8 pixels, $11\,013\,\text{deg}^2$ (inside-64 pixels containing a forest), a median of 471 pixels per forest. Pixel weights are the `picca` weights $w = 1/(\eta\sigma_N^2 + \sigma_{\text{LSS}}^2(\lambda))$ with $\eta \simeq 1$ and $\sigma_{\text{LSS}}^2 = 0.07\text{--}0.19$: the diagonal of the inverse covariance, including the signal variance. The per-forest noise power $P_N = \sigma_N^2 \Delta\chi_{\text{pix}}$ has median $0.54\,h^{-1}\text{Mpc}$ and a scatter of 2.2 in $\ln P_N$.

Tracers. The DR1 large-scale-structure clustering catalogues v1.5 [?]: LRG ($0.4 < z < 1.1$), ELG.LOPnotqso ($0.8 < z < 1.6$) and QSO ($z < 1.75$ here), with their completeness, redshift-failure and imaging-systematics weights, and two random catalogues per Galactic cap ($5000\,\text{deg}^{-2}$) carrying

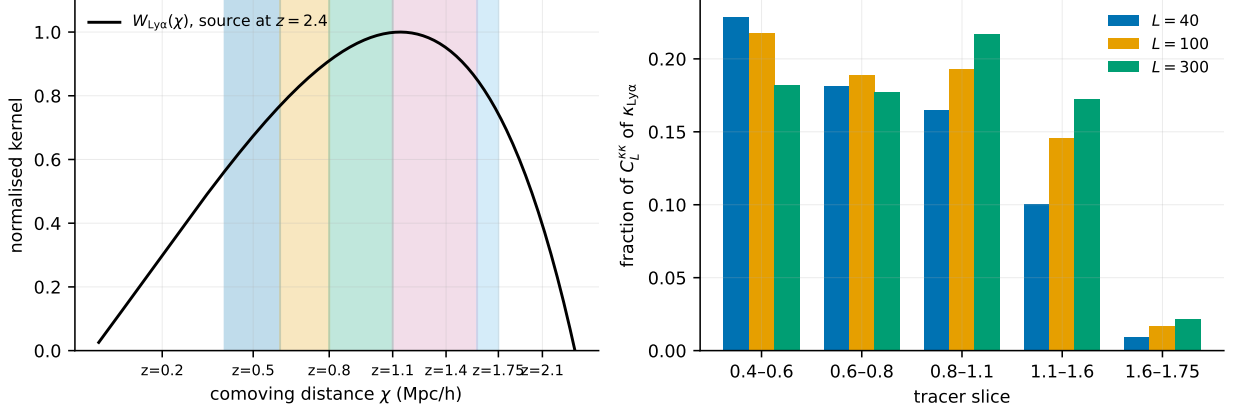


Figure 1: Left: the lensing kernel of a source at $z = 2.4$ (the effective forest redshift) with the five tracer slices shaded. Right: the fraction of the power of $\kappa_{\text{Ly}\alpha}$ contributed by each slice at $L = 40$, 100 and 300 (Limber, non-linear matter power); the slices together hold 69–77 per cent.

redshifts and weights. They are split into five redshift slices, 0.4–0.6, 0.6–0.8 (LRG), 0.8–1.1 (LRG, ELG, QSO), 1.1–1.6 (ELG, QSO) and 1.6–1.75 (QSO), all in front of the forest box ($z < 1.8$). Figure 1 shows where they sit under the lensing kernel of a source at $z = 2.4$: the five slices contain 69–77 per cent of the power of $\kappa_{\text{Ly}\alpha}$ (the forest’s own slab contributes 0.5 per cent; $z < 0.4$ the rest).

CMB lensing. The ACT DR6 baseline convergence [Madhavacheril et al., 2024, Qu et al., 2024] is used only to cross-check the tracer biases (Section 3.3).

3 Estimator

3.1 Lensing of pair correlations

Let $\delta_F(\boldsymbol{\theta}, \chi)$ be the forest flux fluctuation observed along a sightline at angular position $\boldsymbol{\theta}$ and comoving distance χ . Lensing maps the observed position to the true one, $\boldsymbol{\theta}_{\text{true}} = \boldsymbol{\theta} + \boldsymbol{\alpha}(\boldsymbol{\theta})$, with $\boldsymbol{\alpha} = \nabla\phi$ and $\kappa = -\nabla^2\phi/2$, so that the observed field is the true field sampled at $\boldsymbol{\theta} + \boldsymbol{\alpha}$. For a pair of sightlines a, b the correlation of two pixels at transverse separation $r_\perp = \chi|\boldsymbol{\theta}_a - \boldsymbol{\theta}_b|$ and radial separation $r_\parallel$ is the true correlation at the displaced separation,

$$\langle \delta_F(a, \chi_p) \delta_F(b, \chi_q) \rangle = \xi_F(|\boldsymbol{\theta}_{ab} + \boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b|, \chi, r_\parallel) \simeq \xi_F(r_\perp, r_\parallel) + \frac{\partial \xi_F}{\partial r_\perp} \chi \hat{\boldsymbol{\theta}}_{ab} \cdot (\boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b), \quad (1)$$

with $\hat{\boldsymbol{\theta}}_{ab}$ the unit pair direction (from b to a). For a pair separation small compared with the wavelength of the lens, $\boldsymbol{\alpha}_a - \boldsymbol{\alpha}_b \simeq (\boldsymbol{\theta}_{ab} \cdot \nabla) \nabla\phi$: the convergence dilates the transverse correlation isotropically and the shear makes it anisotropic. Equation (1) is exact in the pair displacement and does not require the squeezed limit; the deflection difference is evaluated from the template at the two sightlines.

3.2 Quadratic estimator with a deflection template

Given a template deflection $\boldsymbol{\alpha}^T(\boldsymbol{\theta})$ for the lensing field, the amplitude A of the deflection in the data ($\boldsymbol{\alpha} = A \boldsymbol{\alpha}^T$ in expectation) is estimated from all pixel pairs on all sightline pairs with $3 \leq r_\perp \leq 30 h^{-1} \text{Mpc}$ and $|r_\parallel| < 30 h^{-1} \text{Mpc}$,

$$\hat{A} = F^{-1}(q - \bar{q}), \quad q = \sum_{ab} [\chi_{ab} \hat{\boldsymbol{\theta}}_{ab} \cdot (\boldsymbol{\alpha}_a^T - \boldsymbol{\alpha}_b^T)] \sum_{p \in a, q \in b} w_p w_q \delta_F(p) \delta_F(q) \xi_F'(r_\perp, r_\parallel), \quad (2)$$

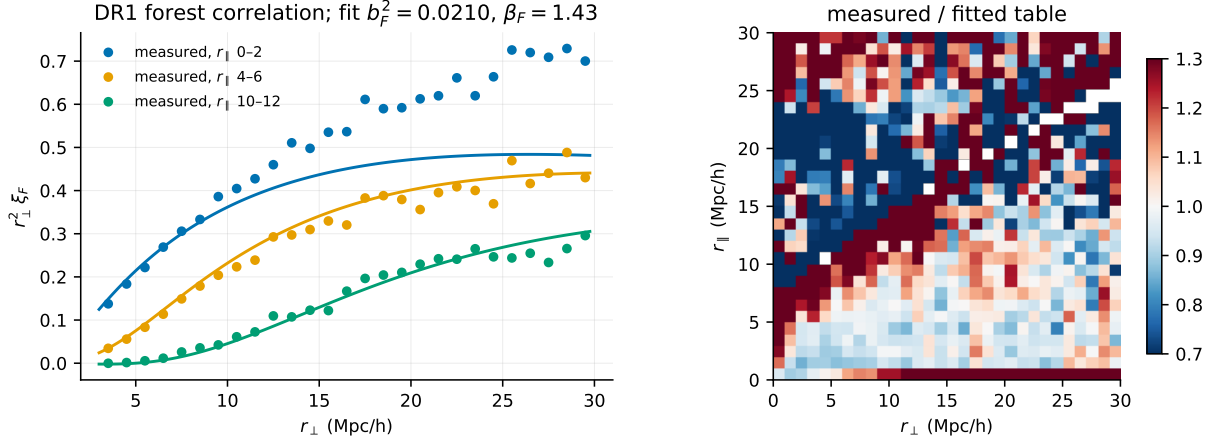


Figure 2: DR1 forest correlation from 1.8×10^8 pixels: measured $r_{\perp}^2 \xi_F$ in three radial-separation strips (points) and the fitted two-parameter table (lines); right, the ratio of the measured $1 h^{-1}$ Mpc cells to the fitted table (the diagonal band is the zero crossing of the anisotropic model). The fit is driven by the small separations that carry the lensing information; at $r_{\parallel} < 2$ and $r_{\perp} > 15 h^{-1}$ Mpc the data exceed the model by 20–40 per cent (Section 6).

with $\xi'_F = \partial \xi_F / \partial r_{\perp}$ the response kernel, $\bar{q}$ the mean field (the same sum with $\delta_F(p)\delta_F(q)$ replaced by ξ_F : it removes the correlation of the sightline density with the template and is not zero on a masked footprint), and F the response matrix, $\sum w_p w_q \xi_F'^2 [\chi \hat{\theta} \cdot \Delta \alpha^T]^2$ up to the same geometric factors. This is the optimal quadratic estimator for the pixel-diagonal inverse covariance; a full per-sightline inverse covariance would shape the weights along $k_{\parallel}$ and we estimate its gain at 8 per cent in signal-to-noise, not worth its complexity here. The pair geometry uses a local tangent-plane metric per pair, valid at the separations used; the sightline density is $19\text{--}22 \text{ deg}^{-2}$ so the pair count is 3.9×10^6 .

The template deflection is split into three multipole bands, $40 \leq L < 100$, $100\text{--}200$ and $200\text{--}300$ (cosine-tapered windows), plus the complementary “junk” band, and each band has a curl partner (the deflection rotated by 90°). The seven components are fitted jointly with the 7×7 response matrix; the science amplitude is the common amplitude of the three gradient bands with the curl and junk amplitudes marginalised (a “curl null” is reported as the mean curl amplitude). Errors are from a HEALPix jackknife on the pair midpoints (inside 8, 301 regions on DR1) and are checked against the Fisher scale $F^{-1/2}$ and against random templates.

The response kernel. $\xi_F(r_{\perp}, r_{\parallel})$ is measured from the same forests in $1 h^{-1}$ Mpc cells and fitted with a two-parameter linear model $P_F(\mathbf{k}) = b_F^2 (1 + \beta_F \mu^2)^2 P(k)$, projected through the per-forest continuum operator (mean and slope removed) and the DESI pixel and resolution windows; the fitted table supplies both ξ_F (mean field) and ξ'_F (kernel). On DR1 the fit gives $b_F^2 = 0.0210$ and $\beta_F = 1.43$ over $3 \leq r_{\perp} < 30$, $r_{\parallel} < 30 h^{-1}$ Mpc (Figure 2). The lensing information is concentrated at small radial separations: 80 per cent of it comes from $r_{\parallel} < 5 h^{-1}$ Mpc and half from $r_{\perp} < 15 h^{-1}$ Mpc (Figure 3).

3.3 Templates from low-redshift tracers

For a tracer with objects at (θ_i, χ_i) and mean density $\bar{n}(\chi)$ per steradian per unit distance, the kernel-weighted map

$$m(\theta) = \sum_{i \in \text{pixel}} \frac{w_i W(\chi_i)}{b \bar{n}(\chi_i) \Omega_{\text{pix}}} - (\text{randoms, scaled}), \quad W(\chi) = \frac{3}{2} \Omega_m \frac{H_0^2}{c^2} (1+z) \chi \frac{\chi_s - \chi}{\chi_s}, \quad (3)$$

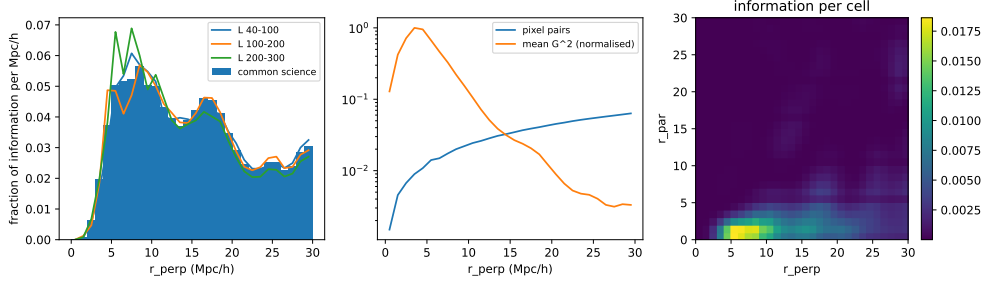


Figure 3: Where the lensing information lives: the cumulative Fisher information of the amplitude versus $r_{\perp}$ and $r_{\parallel}$ on a scale-1 mock, per band and combined. Half of the information is inside $r_{\perp} = 15 h^{-1}\text{Mpc}$ and 80 per cent at $r_{\parallel} < 5 h^{-1}\text{Mpc}$.

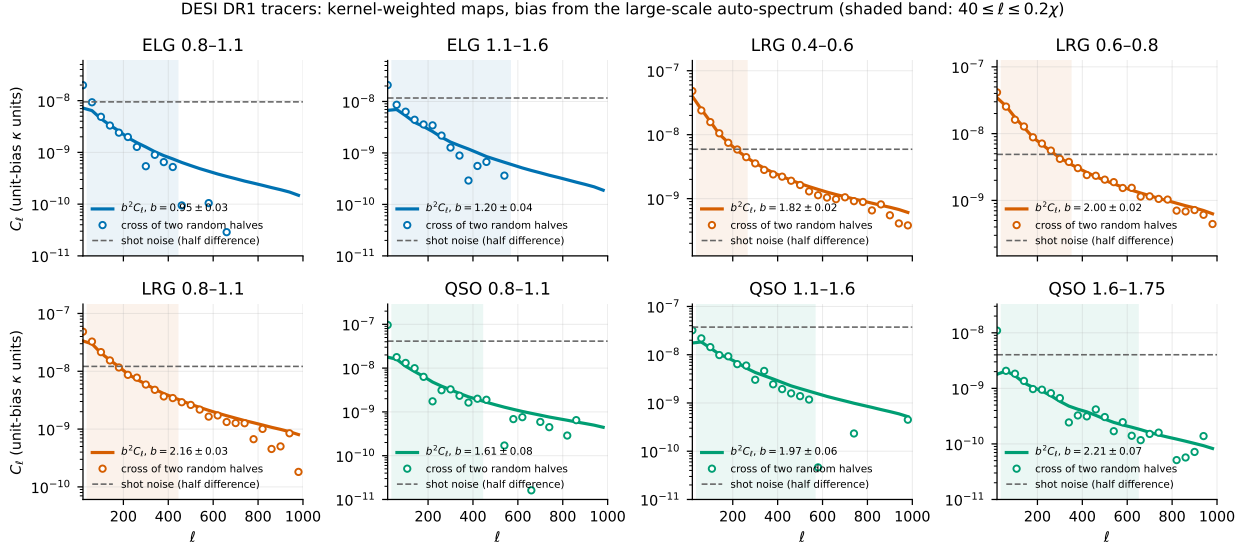


Figure 4: Angular power spectra of the eight DR1 tracer maps in convergence units for unit bias: the cross-spectrum of two random halves (points, no shot noise), the fitted $b^2 C_{\ell}$ (line) and the shot noise measured from the half difference (dashed). The shaded band is the fit range $40 \leq \ell \leq 0.2\chi(z_{\text{mid}})$.

estimates the contribution of the tracer's slice to $\kappa_{\text{Ly}\alpha}$ for a source at $\chi_s = \chi(2.4)$, once the linear bias b of the tracer is known: $\langle m \rangle = \int_{\text{slice}} W \delta d\chi$. The randoms define the footprint (pixels holding at least half the mean random count, which excludes partially covered rim pixels), the completeness (their density smoothed by 1°) and the mean-density subtraction; the map is built at nside 512.

Bias from the angular power spectrum. The bias of every (tracer, slice) is measured from its own map: the cross-spectrum of two random halves of the catalogue (which has no shot noise) is fitted against the Limber spectrum of the slice, $C_{\ell} = b^2 \int_{\text{slice}} W^2 P(k = (\ell + \frac{1}{2})/\chi, \chi) d\chi/\chi^2$, times the pixel window and a Monte-Carlo transfer of the survey mask (0.8–0.9 in the band for the fragmented DR1 footprint), on large scales only, $40 \leq \ell \leq 0.2\chi(z_{\text{mid}})$ ($k \leq 0.2 h \text{Mpc}^{-1}$), with Gaussian bandpower errors. The shot noise of the full map is measured from the half difference and agrees with the weighted Poisson expectation $\sum_i (u_i w_i)^2 \Omega_{\text{pix}}^2 / \Omega_{\text{mask}}$ to 1–3 per cent for the LRG and ELG samples (plain $1/\bar{n}$ is 20–34 per cent too low because the weights enter squared). Figure 4 shows the spectra and Table 1 the biases.

Cross-check with CMB lensing. The cross-spectrum of each unit-bias map with the ACT DR6 convergence map on their overlap (joint sky fraction 0.05–0.08; ACT mask squared on the κ side as the release prescribes; f_{sky} scaling) gives b from $\langle m \kappa_{\text{CMB}} \rangle = b \int_{\text{slice}} W W_{\text{CMB}} P d\chi/\chi^2$ without any

tracer, slice	objects	b (auto-spectrum)	b (ACT κ cross)	ratio	b (literature)
LRG 0.4–0.6	506 911	1.82 ± 0.02	1.50 ± 0.20	0.83	~ 1.9
LRG 0.6–0.8	771 894	2.00 ± 0.02	2.04 ± 0.18	1.02	~ 2.1
LRG 0.8–1.1	859 822	2.17 ± 0.03	2.35 ± 0.17	1.08	~ 2.3
ELG 0.8–1.1	1 016 365	0.96 ± 0.03	0.58 ± 0.13	0.60	~ 1.3
ELG 1.1–1.6	1 415 707	1.20 ± 0.04	1.12 ± 0.14	0.93	~ 1.4
QSO 0.8–1.1	143 846	1.61 ± 0.08	0.95 ± 0.23	0.59	1.7
QSO 1.1–1.6	358 775	1.97 ± 0.06	2.14 ± 0.20	1.09	2.1
QSO 1.6–1.75	115 295	2.22 ± 0.07	2.02 ± 0.40	0.91	2.5

Table 1: Linear biases of the DR1 tracer slices from their large-scale auto-spectra (split-catalogue cross-spectrum, $40 \leq \ell \leq 0.2\chi$) and from the cross-spectrum with the ACT DR6 lensing map; the last column gives the DESI clustering values and, for the quasars, $b_{\text{QSO}}(z) = 0.278[(1+z)^2 - 6.565] + 2.393$ at the slice centre, as orientation.

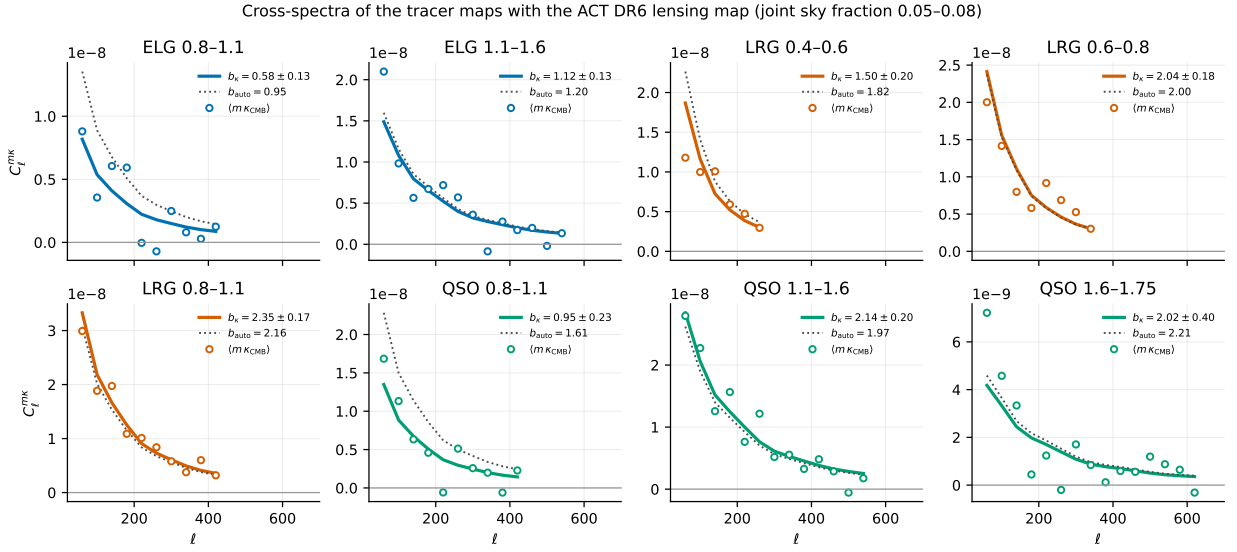


Figure 5: Cross-spectra of the unit-bias tracer maps with the ACT DR6 lensing map (points) with the theory scaled by the cross-correlation bias (solid) and by the auto-spectrum bias (dotted).

shot-noise model (Figure 5, Table 1). Every slice is detected in cross-correlation at $5\text{--}13\sigma$; six of the eight biases agree with the auto-spectrum values within $1\text{--}2\sigma$ (10–17 per cent), while the ELG and QSO of the 0.8–1.1 slice are 40 per cent lower in the cross (2.9σ each), the ELG auto-spectrum also showing the falling $b(\ell)$ characteristic of residual imaging systematics. The templates use the auto-spectrum biases.

Wiener combination and the deflection template. Within a slice the tracer maps (in convergence units, $m/\hat{b}$) are combined with the weights $w_\ell = C_\ell^{-1} s_\ell$ of the model covariance $C_{kl} = S_\ell + N_k \delta_{kl}$ (S_ℓ the slice’s convergence spectrum, N_k the measured shot noise): inverse-shot-noise weighting of the tracers, Wiener-suppressed where noise dominates. The slices are independent, so the sum of the filtered slice maps is the conditional expectation of $\kappa_{\text{Ly}\alpha}$ given all tracers; its deflection, $\alpha^T = \nabla\phi$ with $\phi_{LM} = 2\kappa_{LM}/[L(L+1)]$, evaluated at the sightlines (`healpy` derivatives at `nside 512`), is the combined template. Because the filter is the conditional-expectation coefficient, the amplitude estimator of Section 3.2 is normalised to $A = 1$ for the true lensing without further calibration; each slice is also fitted on its own and the five slice amplitudes are combined with their joint jackknife covariance as a cross-check.

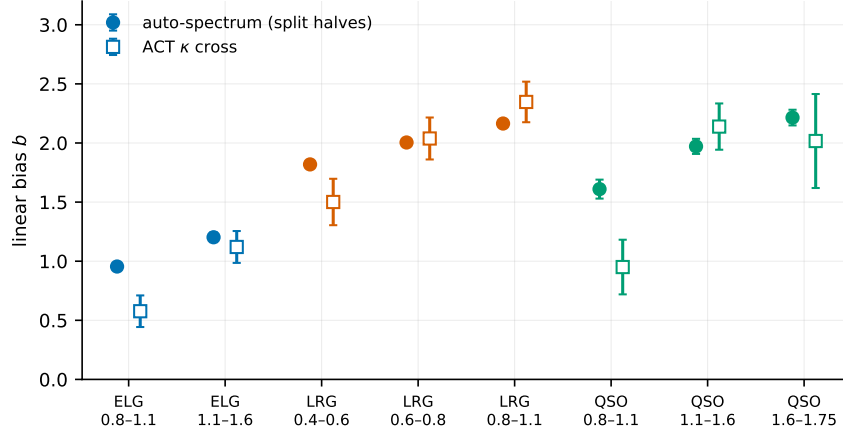


Figure 6: The two bias estimates of Table 1.

3.4 Why not a CMB-lensing template

The same estimator cross-correlated with κ_{CMB} measures lensing plus the intrinsic forest–forest–density three-point function of the forest’s own redshift range. Its linear part, the response of the small-scale forest power to long-wavelength modes, can be removed with a kernel-matched template built from the $z > 2$ quasars (the “deprojection” of the main report), but the second-order part, $\langle m_p m_q T \rangle$ with m the long-wavelength component of the forest and T a non-Gaussian tracer template, cannot be removed by one coefficient: on 400 mock realisations it biased the deprojected amplitude by -0.41 ± 0.18 with no lensing and no response present. The lensing signal, the response and this term all act through the same long modes at the pair separations used, so the term has to be modelled, not filtered. A tracer at $z < 1.8$ shares no density modes with the forest at $z > 2.1$: its correlation with the pair products is lensing only, plus the magnification of the sightline quasars (absorbed by the mean field) and shared sky systematics.

4 Validation on mocks

The mocks are flat-sky FFT realisations of a $20^\circ \times 20^\circ$ patch: a Gaussian matter field on a $2 h^{-1}\text{Mpc}$ grid over the forest box $2.1 < z < 3.0$ (plus margins) from which the forest is a linear, redshift-space-distorted filter with $b_F = -0.13$, $\beta_F = 1.6$ and a small-scale cut-off; 22 sightlines per deg^2 with pixel noise $P_N = 0.33 h^{-1}\text{Mpc}$ (log-normal scatter across forests) and per-forest continuum projection; the ACT DR6 mask cut-out and a smooth completeness pattern; lognormal quasars behind the slab draw the sightlines. The foreground of $\kappa_{\text{Ly}\alpha}$ (and of κ_{CMB}) is generated in the five tracer slices as Gaussian projected-density maps with the exact Limber covariance between the density and the two convergence contributions, plus the uncovered rest; each slice hosts 2-D lognormal Poisson tracers of the projected density smoothed transversely by $3 h^{-1}\text{Mpc}$ (a lognormal of the raw $0.7 h^{-1}\text{Mpc}$ cells is absurdly non-Gaussian), with DR1-like biases and densities. The forest is lensed by the full $\kappa_{\text{Ly}\alpha}$ (sampling at the displaced positions), and every realisation carries both a lensed and an unlensed forest.

The protocol (GATES.md v7) was frozen before the scale-1 campaign: 400 fresh realisations, templates built from each realisation’s own tracers exactly as on the data (bias from the auto-spectrum, Wiener combination, sum over slices), and the gate “an unbiased detection against a lognormal redshift tracer”. Figure 7 and Table 2 give the result: the combined null and recovery are unbiased, the paired response (the same realisation lensed and unlensed, free of sample variance) is 0.966 ± 0.025 , equal to the response of the truth template (0.970 ± 0.018 ; the 3 per cent deficit of the noisy sparse sample was already seen with the CMB template and is a property of the noisy forest, not of the template chain), the per-slice responses are unity within errors, a template from another realisation gives zero, the curl is zero, the jackknife error matches the ensemble scatter,

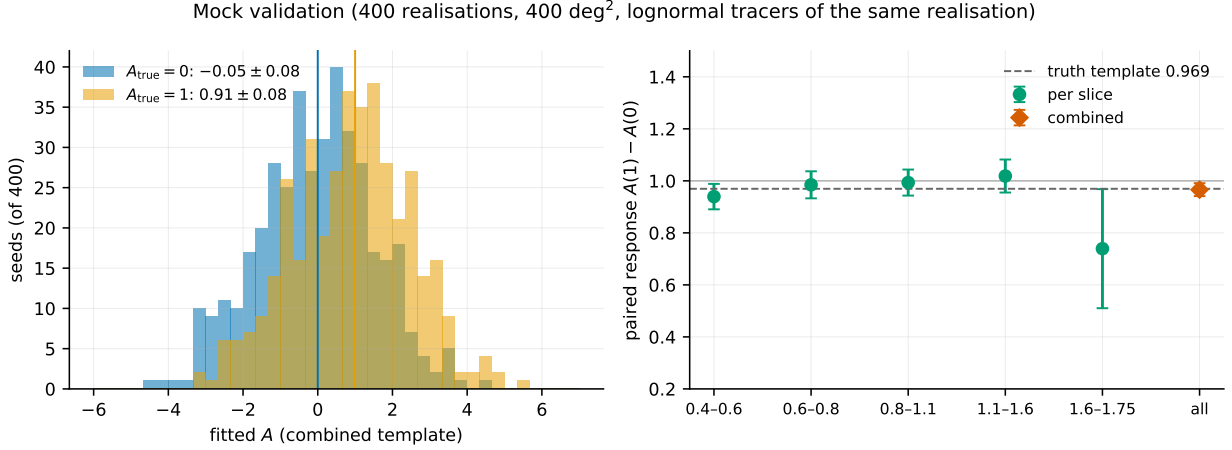


Figure 7: Mock validation on 400 realisations. Left: the combined-template amplitude for an unlensed ($A_{\text{true}} = 0$) and a lensed ($A_{\text{true}} = 1$) forest of the same realisations. Right: the paired response $A(1) - A(0)$ per slice and combined, against the truth template.

statistic (400 realisations, 400 deg ²)	value	requirement
combined null, $A_{\text{true}} = 0$	-0.05 ± 0.08	$ \cdot \leq 2 \text{ SEM}, 95\% \text{ bound} \leq 0$
combined recovery, $A_{\text{true}} = 1$	0.91 ± 0.08	$ \cdot - 1 \leq 2 \text{ SEM}, 95\% \text{ bound} \leq 1$
paired response $A(1) - A(0)$	0.966 ± 0.025	$ \cdot - 1 \leq 2 \text{ SEM}$
truth-template paired response	0.970 ± 0.018	report
per-slice responses (0.4–0.6 ... 1.6–1.75)	0.94, 0.99, 0.99, 1.02 (± 0.05); 0.74 ± 0.23	each within 2 SEM
other-realisation template	0.01 ± 0.08	$ \cdot \leq 2 \text{ SEM}$
curl (combined)	-0.07 ± 0.10	$ \cdot \leq 2 \text{ SEM}$
scatter / RMS jackknife	0.96	0.7–1.3
normalisation slope on $A \in \{0, \frac{1}{2}, 1, 2\}$	0.97 ± 0.06	within 2 SEM of 1
response on – off	0.08 ± 0.09	$ \cdot \leq 2 \text{ SEM}$
injection expectation, odd slope at $ A = 0.25$	0.980	1 ± 0.05
tracer biases $b_{\text{fit}}/b_{\text{true}}$	$0.97\text{--}1.01$ ($\pm 0.002\text{--}0.004$)	2 SEM (fails for 6 of 8)
per-seed scatter of the combined amplitude	1.56	(CMB deprojected estimator:

Table 2: Iteration-7 acceptance (`report/lowz_validation.md`).

and the noise-free injection expectation (the bookkeeping test of signs, cuts and band basis) has odd slope 0.98. The only rows of the protocol that fail are the tracer-bias precision rows: the auto-spectrum bias of a lognormal tracer differs from the generator’s by 1–3 per cent, resolved at $N = 400$ against a 2-SEM rule that amounts to 0.4 per cent; the effect on A is the same 1–3 per cent per slice.

5 Measurement on DR1

The full run (one Perlmutter node, 24 minutes, 15 GB) fits the forest correlation, builds the pair catalogue (3.9×10^6 sightline pairs), evaluates the five slice templates and the combined template at the 373 756 sightlines, and fits the seven-component basis. Table 3 and Figure 8 give the result:

$$A = 0.01 \pm 0.67 \quad (\text{combined template, jackknife; Fisher scale } 0.60), \quad A < 1.11 \text{ at } 95 \text{ per cent (one-sided)}. \quad (4)$$

The five slices are individually consistent with zero and their jackknife-covariance combination, 0.00 ± 0.64 , agrees with the combined template (the slice correlations are below 0.08). The nulls: the curl amplitude is -0.18 ± 1.79 ; forty Gaussian random realisations of the combined map’s own spectrum through the tracer mask, fitted exactly like the data, give a mean of -0.13 ± 0.10 and a scatter of 0.64 against the RMS jackknife error 0.73 (the jackknife is 10 per cent conservative);

template	A	jackknife	Fisher	curl
combined (five slices)	0.010	0.670	0.601	-0.18 ± 1.79
slice 0.4–0.6	−0.62	1.09	1.13	-2.3 ± 3.4
slice 0.6–0.8	−0.37	1.37	1.08	1.9 ± 3.1
slice 0.8–1.1	1.30	1.19	1.11	2.4 ± 3.4
slice 1.1–1.6	−1.07	1.78	1.65	-5.5 ± 4.7
slice 1.6–1.75	5.7	6.1	6.0	-9 ± 18
jackknife combination of the slices	0.002	0.638		
random templates (40)	-0.13 ± 0.10	scatter 0.64	RMS jackknife 0.73	

Table 3: DR1 lensing amplitude, single slab $2.1 < z < 3.0$ (`report/stageb/dr1_lowz.md`).

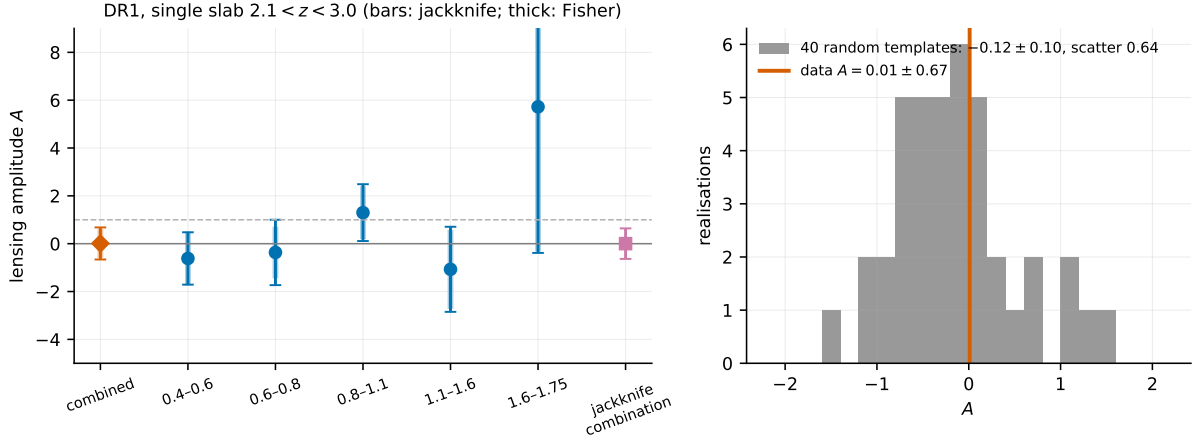


Figure 8: Left: the DR1 amplitudes per slice and combined (thin bars: jackknife, thick: Fisher) and the jackknife-covariance combination of the slices; the dashed line is the Λ CDM value. Right: the forty random-template amplitudes with the data value.

the noise-free injection expectation of the combined template’s own deflection on the real sightlines has odd slope 1.014. The combined science bands are 0.07 ± 1.15 , 0.33 ± 1.04 and -0.64 ± 1.38 for $L = 40\text{--}100$, $100\text{--}200$ and $200\text{--}300$.

6 Caveats and systematics

At an error of 0.67 on A , systematic effects of 20 per cent are irrelevant; the following are recorded for the full-DESI measurement.

- *Tracer biases.* The auto-spectrum and CMB-cross biases agree to 10–17 per cent for six tracers; the ELG and QSO of the 0.8–1.1 slice disagree by 40 per cent. A bias error f scales that slice’s amplitude by $1/f$; with the slice weights the combined amplitude moves by a few per cent. The ELG auto-spectrum has a scale-dependent bias above $\ell \simeq 300$ (imaging systematics).
- *Forest correlation model.* The two-parameter Kaiser table is fitted over $3 \leq r_{\perp} < 30 h^{-1}\text{Mpc}$ with pair-count weights; at $r_{\parallel} < 2$ and $r_{\perp} > 15 h^{-1}\text{Mpc}$ the data exceed it by 20–40 per cent (Figure 2), and the first radial bin shows the excess correlation of pixels at the same wavelength on different sightlines (sky and calibration residuals). Both enter the kernel and the mean field; the mock normalisation (0.97) does not include them. A model with Arinyo-i-Prats non-linear terms and a same-wavelength nuisance is the obvious refinement.
- *Continuum projection.* The kernel’s projection through the per-forest mean-and-slope removal is geometry-averaged (one representative forest); DR1 forests vary in length. The injection

expectation on the real geometry (1.014) tests the bookkeeping of this operator but not its physical accuracy.

- *Mask coupling.* The tracer auto-spectra use a Monte-Carlo mask transfer; the CMB cross uses the f_{sky} approximation (10–15 per cent), which the ACT simulations would refine.
- *Magnification and selection.* The sightline quasars are magnified by the foreground; the mean field absorbs the resulting modulation of the pair density, as in the mocks, where the sightline draw carries the magnification.
- *Not yet done.* Tomographic sub-slabs of the forest, the Planck PR4 cross-check of the biases, the propagation of the bias uncertainties into the error.

7 Conclusions

A pair-based quadratic estimator of the forest’s lensing deflection, projected on a template from DESI galaxies and quasars at $z < 1.75$, is unbiased on 400 self-lensed mocks with lognormal tracers and gives $A = 0.01 \pm 0.67$ on the DR1 forest, with clean curl, random-template and injection tests. The template-side biases are measured internally and confirmed by the ACT lensing cross-correlation. The precision corresponds to a 1.5σ deficit from the Λ CDM value, as forecast for DR1; the same pipeline on the complete DESI sample (roughly twice the forests and deeper coverage at $z > 2.5$) should reach $2\text{--}3\sigma$, and the tracer template can grow with the BGS at $z < 0.4$ and with the full-depth ELG sample.

References

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